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MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented ☐ Application oriented ☐ Product oriented ☐ Other (please specify) Specify if others:	Batch ID:	B3

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Title of the Project: AI-LLMs-Agent based LLM for web application Development

Abstract

As Large Language Models (LLMs) increasingly dominate the software development landscape, there remains a significant gap between basic code snippet generation and the autonomous creation of fully functional, production-ready web applications. This project introduces an Intelligent Web Assistant powered by a Multi-Agent Hierarchical Architecture designed to automate the end-to-end development lifecycle across diverse application domains.

The proposed system moves beyond simple prompt-response interactions by implementing a Stateful Orchestration Layer that manages a persistent project directory. Central to the architecture is an Agentic RAG (Retrieval-Augmented Generation) framework, which allows a specialized Knowledge Agent to dynamically retrieve real-time technical documentation and "gold-standard" code patterns from a vector database (ChromaDB), ensuring the system remains current with modern frameworks like React and Tailwind CSS.

A key innovation of this project is the Autonomous Self-Healing Loop. This involves a Reviewer Agent that executes generated code within a secure sandbox environment (E2B/Docker), captures runtime errors, and triggers iterative debugging cycles without human intervention. By utilizing LangGraph to manage complex reasoning loops and Gemini 3.7 Flash for high-context project oversight, the assistant effectively transforms natural language intent into a live-previewed, multi-file web application. This research demonstrates the potential of multi-agent systems to serve as comprehensive software engineering partners, significantly reducing the manual overhead in web development.

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References (Category specific)

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2. **Large Language Model-Based Agents for Software Engineering: A Survey (2024)**
   https://arxiv.org/abs/2409.02977
3. **Agents in Software Engineering: Survey, Landscape, and Vision (2024)**
   https://arxiv.org/abs/2409.09030
4. **From LLMs to LLM-based Agents for Software Engineering: A Survey of Current, Challenges and Future (2024)**
   https://arxiv.org/abs/2408.02479
5. **MetaGPT: Meta Programming for Multi-Agent Collaborative Framework (2023)**
   https://arxiv.org/abs/2308.00352
6. **Benchmarking and Studying the LLM-based Agent System in End-to-End Software Development (2025)**
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Project Guide

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